

FatTail Labs — IKI Product System Design v0.1

Status: DRAFT. Not BUILD AUTHORITY, not a plan. High-level design against `DRAFT-FatTail-Labs-IKI-Product-System-Spec-v0_1.md`.

Filename: provisional — Coach names it.

Altitude: the shape of the system, not the detail of any part. Enough that the pieces can be argued about and the boundaries can be checked; not enough to build from. Every open in the spec's §10 stays open here.

Scope statement (DL-539)

Active program: IKI Product System.

Files / trees this document touches: none. Design only — no code, no routes, no schema.

Touches outside program: NONE.

Advisor honesty, prominent. This design is written **without a disk read**. Nobody has looked at how the runner loads a template today, what a template's contract actually contains, how the help system attaches to a feature, or what the wiki agent expects when something declares intent. Every place that matters is marked **[unverified]**. Where this design and the tree disagree, **the tree wins** — the pattern has already corrected this program four times this week (the envelope, the marks tape, the flat day, the launchd plist).

1. The shape in one paragraph

A **product** is a constellation, not a template. The **Factory** makes constellations with agents pulling work; a human promotes them. **IKI Lab** is where a member finds one, tries it in the place it will live, buys it, and arranges it on a desk. The **Runner** is an execution environment that loads products, gives each one a context, and keeps them from breaking each other. Everything a member can click is a thing that will work, because the system refuses to offer what it cannot deliver.

2. The object model

Six objects. Everything else is a relationship between them.

Object	What it is	Notes
Part	One runnable product. Declares what it needs, what it produces, and what shape it requires.	§2.1
Arrangement	A set of parts in a layout. Made by someone, so it has an author and a version.	Spec §14.3 — whether this is a <i>product</i> is Coach's
Constellation	A part plus its guide, embedded help, wiki page and landing page. This is what ships.	Spec §5
Context	What a running part is given: data, clock, surfaces. One per Runner instance.	§4
Card	A unit of Factory work. Carries its own state in language a human reads.	§6
Grant	What a member holds that lets a part run for them.	Spec §14.9 — commerce shape is Coach's; this design only needs <i>something</i> to exist

2.1 A part's contract

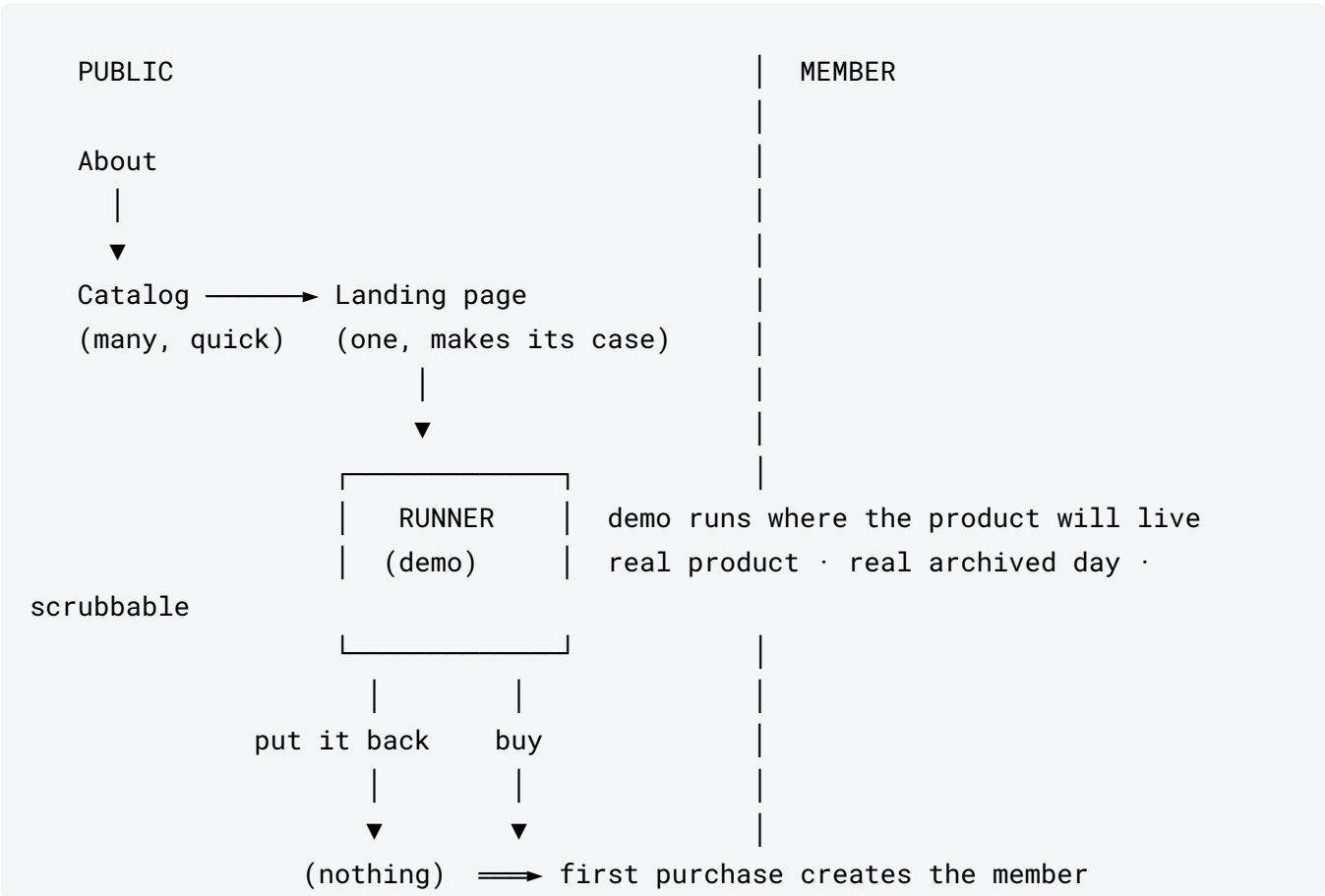
The contract is what makes a part a product rather than a function the host happens to call. Four declarations:

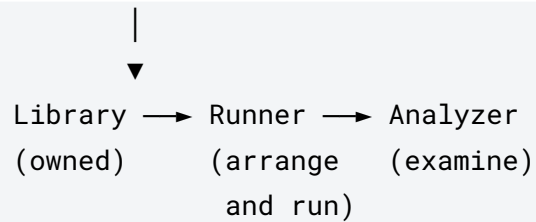
Declaration	Why it exists
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Needs	What data and services. A chain generation, marks, the archive, the playhead. The environment satisfies it or refuses to load the part.
Produces	What it renders, and — for the REST and pipeline paths — what it returns. Without this a part can only paint.
Shape	Fully responsive · minimum size · locked aspect ratio (spec §0.4).
Identity	Name, version, author, credentials.

The whole system leans on this. A declared contract is what lets a layout be validated before it is saved, a landing page be generated, a part be called over REST, and a demo be run without special-casing. **[unverified]** — today's templates almost certainly declare far less than this, and the gap between what they declare and what this needs is the single largest unknown in the design.

3. The member funnel





The membership boundary sits at the first purchase, not at signup. Everything left of it is public: about, catalog, landing pages, and a demo that is the real product on real data.

Three consequences of that boundary.

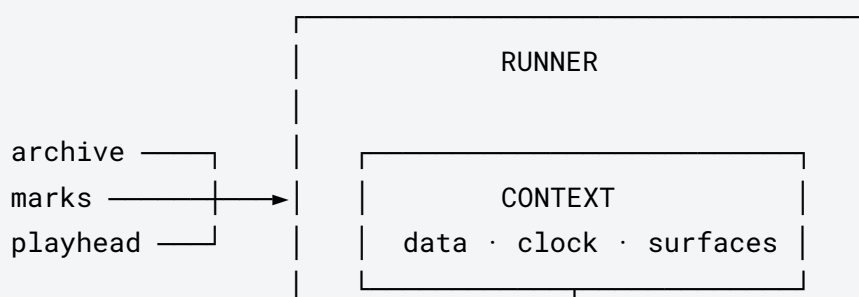
The demo is the trial, so there is no trial period and nothing expires. A member pays when they want to keep something on their desk.

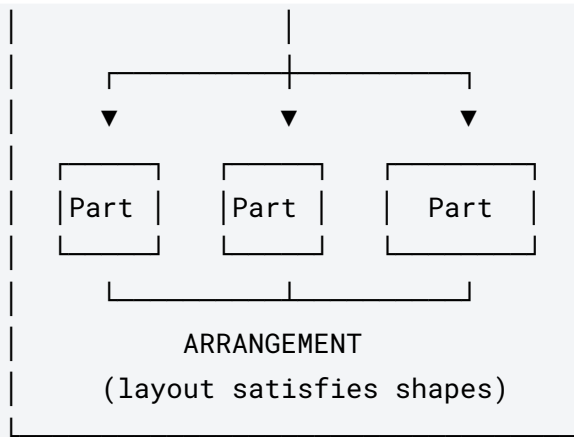
The catalog and landing pages are public and indexable — the same shape as catalog and course pages under the existing SEO layer. Landing pages are an acquisition surface, not merely a purchase step.

And the archive serves a public path for the first time. A demo pulls an archived day into an unauthenticated browser. Every archive route currently requires a bearer, deliberately. This is spec §14.12 and it is the most consequential open in the funnel.

4. The Runner as an execution environment

The claim in spec §0.1 is that this is an operating system for products. That claim is only true if the environment does four things a template host does not.





Parts read only from the context. There is no second source to read.

Load and unload. A part is installed, versioned, and removable. A template that only ever exists because the host imported it cannot be bought or removed.

One context, and nothing else reachable. Every part reads the same data and the same clock. This is not a convention — it is the Time Machine pattern, where the live source is *structurally unreachable* under a playhead rather than merely unused. With several parts on one surface, that stops being tidiness and becomes the isolation rule: a part cannot reach something it was not given, so it cannot break another part by reading the wrong thing. **Spec §14.4 — Coach rules whether this is the model.**

Layout satisfies declarations. The arrangement is not pixels; it is a set of parts whose shape declarations are satisfied. That is what lets an arrangement be validated before saving, refuse rather than render something unusable, and travel between a laptop and a monitor while staying recognisably the same desk.

A failing part is a named hole, not a broken dashboard. One part that cannot load says so in its own cell. The desk does not go down.

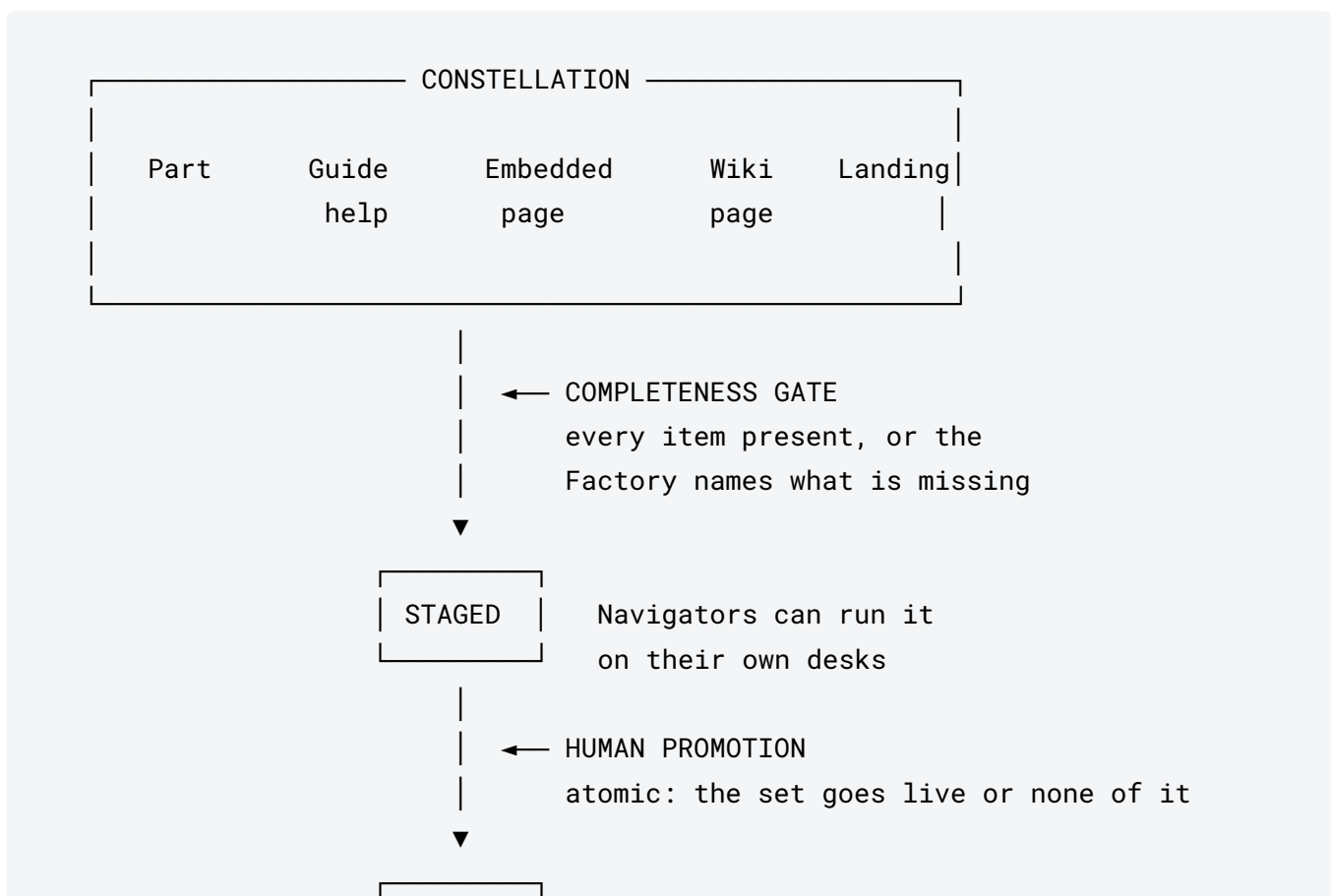
4.1 Temporary parts

A demo part and a bought part are the same code in the same environment. What differs is that a demo is temporary, and the platform already has a settled pattern for temporary things — rehearsal objects.

Badged	Visibly not owned. A member is never unsure which parts they have bought.
Isolated	No alerts, no trade log, no persistence.
Disposed with an announcement	<i>Put it back</i> removes it and says so. Nothing vanishes silently.
Non-destructive	A saved arrangement is restored exactly, or the demo occupies a slot that is clearly temporary.

This is the third use of that pattern — rehearsal objects, then dark positions, now demos. Worth naming as a general rule rather than rediscovering it a fourth time: *a temporary thing is badged, isolated, and disposed with an announcement.*

5. The constellation and the promotion gate



```
| PUBLIC | one-way. never pulled back.
```

The gate is before staging, not after. Staging receives a whole constellation; it is not where one is assembled. That is spec §0.17 and it is the load-bearing sequencing decision in the Factory.

Completeness is checkable, not asserted. The Factory says a product is not stageable **and names what is missing** — the same discipline as a calendar naming the day it cannot load. This puts the wiki agent on the critical path: a product waiting on its wiki entry is a product not shipping. **[unverified]** — whether the wiki agent can be asked *is this done* is unknown and it decides whether the gate is real or aspirational.

Staging is a limited-visibility release, not a rehearsal area. Navigators run a finished thing on their own arrangements, which finds what a checklist does not.

Promotion is atomic and human. Atomic because a product live with no help, or a wiki page pointing at something unpurchasable, is a half-assembled constellation — the same shape as the one-space transaction rule. Human because agents can prove every card is closed but cannot judge whether this should exist in public under Coach's name.

6. The Factory

```
BACKLOG → RESEARCH → BUILD → [gate] → STAGING → [human] → STORE
```

```
agents with skills  ┌  
                    └─ pull.  nothing is assigned.  
humans with roles   └
```

First incarnation: Kanban with pull. That is what exists and it works.

The relationship to Spaces, which Coach has already ruled. The Factory is the first Spaces instance, admin-only, with the lanes dying. Those reconcile rather than collide: **Spaces replaces the lanes, not**

the pulling. Pull is already entity behaviour, so the working method does not change when the substrate does.

Two disciplines that cost nothing now and keep the later swap cheap:

1. **Nothing addresses a specific agent.** A card says what it is, never who works it. Addressing surviving the death of the lanes is Coach's own lie-check for the Spaces proof.
2. **A card carries its own state in readable language,** not only in the column it sits in. If the column is the only thing that knows a card has been researched, the column cannot be removed.

And the completeness gate stops being a checklist under Spaces. A constellation is complete when every card in its set is claimed, worked and released — visible in the space rather than needing a separate list. The gate and the substrate become the same fact seen twice.

7. Delivery paths

Three ways to consume one product. **This is where the contract earns its keep** — a part that only paints cannot be called.

Path	What it needs from the contract
Desk	Needs · shape. The Runner satisfies and places it.
REST	Needs · produces. No surface; inputs in, outputs out.
Pipeline	Needs · produces · identity. Wired into something else, so version matters.

Advisor: if a part's *produces* is declared for the desk path anyway, REST is close to free and pipeline is REST with versioning discipline. If it is not declared, all three are separate implementations. **Spec §14.6 — whether all three are v1 is Coach's**, but the contract decides whether that question is expensive or cheap.

8. Boundaries

Where this system stops, so it is not designed as though on empty ground.

Boundary	
Options Lab	Provides the archive, the clock, and the existing template surface. Not changed by this system.
Time Machine	The demo's data and clock. Spec §14.10 — whether the Runner shares its playhead.
Practice	Supplies the symbol universe a product runs against.
Help system	Reused as-is (§0.15). The existing build-time obligation keeps guides from drifting.
Wiki + Oscar	Authors the wiki page from a declaration of intent. Producers do not learn the wiki's shape.
Identity / payments	Not entered. First purchase creates a member (§0.14); how is Coach's and Mike's.
SEO layer	Landing pages are public and indexable, same shape as existing catalog pages.
No-profit-claims	Applies to every landing page. Products sell on capability and process, never outcome. Tango gates it.

9. What has to be read before this becomes a plan

The four things marked **[unverified]**, in the order they matter:

1. **What a template declares today.** The gap between that and §2.1's contract is the largest unknown in the design and it determines the size of the whole program.
2. **How the Runner loads a template now.** Whether load/unload/version is a small addition or a

rewrite.

3. **Whether the wiki agent can answer *is this done*.** Decides whether the completeness gate is real or aspirational.
4. **How the help system attaches to a feature.** Decides whether per-product help is reuse or new work.

A disk read on those four turns this design from plausible into checkable. Nothing should be planned before it.

10. Open

Every open in the spec's §10 stands, untouched. This design adds none and decides none.

Two it makes sharper rather than answering:

- **§14.3 (parts and arrangements as two classes).** The design needs both objects to exist regardless; whether the arrangement is *sellable* is a commerce question, not a structural one.
 - **§14.4 (isolation model).** The one-context pattern is what makes the operating-system claim true rather than aspirational. If Coach rules otherwise, §4 changes materially.
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11. Document control

Version	Date	Notes
v0.1 DRAFT	2026-08-29	High-level design against the spec. Six objects; the part contract as the load-bearing declaration; four diagrams — member funnel, Runner execution environment, constellation and promotion gate, Factory pull. Temporary-part rules named as the third use of an existing pattern. Delivery paths shown to depend on the contract. Four unverified facts named as blocking a plan. No opens decided. Not BUILD AUTHORITY.

One-line law:

A part declares what it needs, produces and requires; the environment satisfies it or refuses; the constellation is complete or it is not staged; a human promotes it; and everything a member can click is a thing that will work.